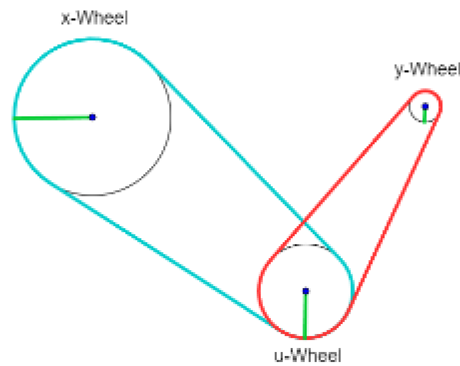
3.5 The Chain Rule

Motivation:
We wish to determine how to take the derivative of a composition of functions in multiple dimensions, i.e., what does chain rule look like for functions beyond $f(g(x))$?

Exploration:
Consider the mechanism to the right. Suppose that every time the x-wheel spins one time, the u-wheel spins 2 times. Furthermore, every time the u-wheel spins 1 time, the y-wheel spins 3 times. Question: If the x-wheel spins one time, how many times does the y-wheel spin? [1]



The y-wheel will spin 6 times for every 1 time the x-wheel spins. If we notate the ratio of change in the u-wheel to change in the x-wheel as $\frac{du}{dx}=\frac{2}{1}$ *(recall that "d" doesn't mean "derivative," it means "infinitesimal difference/change in…"). Similarly, the change in the y-wheel compared to the change in the u-wheel as* $\frac{dy}{du}=\frac{3}{1}$*, we can see that* $\frac{du}{dx}\cdot\frac{dy}{du}=\frac{2}{1}\cdot\frac{3}{1}=\frac{6}{1}=\frac{dy}{dx}$. *This is the basis of chain rule.*

Recall: In calculus one you learned that if $y=f(x)$ and $x=g(t)$ where f and g are differentiable functions, then y is indirectly a differentiable function of t and $\frac{dy}{dt}=\frac{dy}{dx}\cdot\frac{dx}{dt}$.

Note: We will see in this section that the chain rule generalizes nicely to functions of several variables.

Theorem (Case 1 of the Chain Rule):
Suppose that $z=f(x,y)$ is a differentiable function of x and y where $x=g(t)$ and $y=h(t)$ are both differentiable functions of t. Then z is a differentiable function of t and

$$\frac{dz}{dt}=\frac{\partial f}{\partial x}\frac{dx}{dt}+\frac{\partial f}{\partial y}\frac{dy}{dt}$$

Proof: See pages 949 – 950 & Appendix D of the text.

Note: One may see the above equation written as: $\frac{dz}{dt}=\frac{\partial z}{\partial x}\frac{dx}{dt}+\frac{\partial z}{\partial y}\frac{dy}{dt}$.

[1] The corresponding applet can be found here:
https://webspace.ship.edu/msrenault/geogebracalculus/derivative_intuitive_chain_rule.html

Example 1: Suppose that $z = \cos(x+4y)$, $x = 5t^4$, and $y = \frac{1}{t}$. Assuming $t \neq 0$, find $\frac{dz}{dt}$.

By the chain rule we note that:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt} = -\sin(x+4y)(20t^3) + -4\sin(x+4y)\left(\frac{-1}{t^2}\right)$$

$$= -20t^3\sin(x+4y) + \frac{4\sin(x+4y)}{t^2}.$$

It is unnecessary to give an expression purely in terms of *t*. This is because given any value of *t* we can easily evaluate *x* and *y*.

With the generalized chain rule, we will be able to attack more difficult related rates problems.

Example 2: The pressure *P* (in kilopascals), volume *V* (in liters), and temperature *T* (in kelvins) of a mole of an ideal gas are related by $P = \frac{8.31T}{V}$. Find the rate at which the pressure is changing (rounded to the nearest thousandth) when the temperature is 300K and increasing at a rate of 0.1 K/s and the volume is 100 L and increasing at a rate of 0.2 L/s.

We think of both *V* and *T* as functions of *t* (time). By the chain rule we see that:

$$\frac{dP}{dt} = \frac{\partial P}{\partial T}\frac{dT}{dt} + \frac{\partial P}{\partial V}\frac{dV}{dt} = \frac{8.31}{V}\frac{dT}{dt} - \frac{8.31T}{V^2}\frac{dV}{dt}$$

Then, we see that the answer is: $\left.\frac{dP}{dt}\right|_{T=300,\ V=100,\ \frac{dT}{dt}=0.1,\ \frac{dV}{dt}=0.2} = \frac{8.31}{100}(0.1) - \frac{8.31(300)}{100^2}(0.2) \approx -.042\ \text{kPa/s}$.

Theorem (Case 2 of the Chain Rule):
Suppose that $z=f(x,y)$ is a differentiable function of *x* and *y*, where $x=g(s,t)$ and $y=h(s,t)$ are differentiable functions of *s* and *t*. Then, $\dfrac{\partial z}{\partial s}=\dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial s}+\dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial s}$ and $\dfrac{\partial z}{\partial t}=\dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial t}+\dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial t}$.

__Example 3:__ Suppose that $z=y\left(2^x\right)$ where $y=s^2+st$ and $x=e^{st}$. Find the exact value of $\dfrac{\partial z}{\partial s}$ when $s=2$ and $t=0$.

First, we note by case 2 of the chain rule that $\dfrac{\partial z}{\partial s}=\dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial s}+\dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial s}=(\ln(2))\,y\left(2^x\right)\left(te^{st}\right)+2^x(2s+t)$.
We note that when $s=2$ and $t=0$, then $y=4$ and $x=1$.
Thus, $\left.\dfrac{\partial z}{\partial s}\right|_{s=2,\ t=0}=(\ln(2))4\left(2^1\right)(0)+\left(2^1\right)(4+0)=8$.

Note: It is important to see that the symbol ∂z does not have a meaning in and of itself (unlike dz). Therefore, one should recognize that $\dfrac{\partial z}{\partial s}=\dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial s}+\dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial s}\neq\dfrac{\partial z}{\partial s}+\dfrac{\partial z}{\partial s}$ (unless it is the case that $\dfrac{\partial z}{\partial s}=0$).

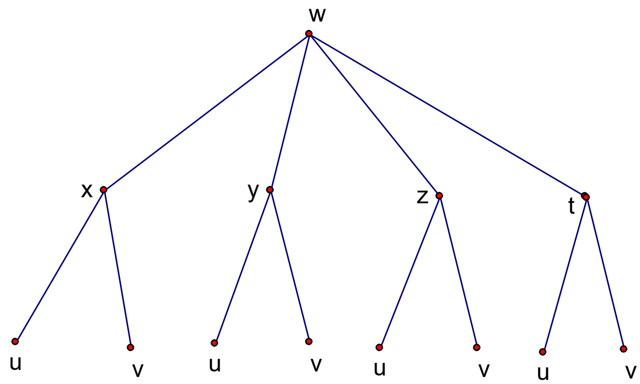
Theorem (General Chain Rule):
Suppose that *u* is a differentiable function of the *n* variables $x_1,x_2,\ldots,x_n$ and each x_j is a differentiable function of the *m* variables $t_1,t_2,\ldots,t_m$ and $\dfrac{\partial u}{\partial t_i}=\dfrac{\partial u}{\partial x_1}\dfrac{\partial x_1}{\partial t_i}+\dfrac{\partial u}{\partial x_2}\dfrac{\partial x_2}{\partial t_i}+\cdots+\dfrac{\partial u}{\partial x_n}\dfrac{\partial x_n}{\partial t_i}$ for each $i=1,2,\ldots,m$.

Proof: The proof the case 2 of the chain rule and the general chain rule are beyond the scope of the course.

Visualizing the General Chain Rule:

A great way to think through the general chain rule is with the help of a tree diagram. To make the idea of the tree diagram concrete, I will use a simple example.

Suppose that $w=f(x,y,z,t)$ and $x=x(u,v)$, $y=y(u,v)$, $z=z(u,v)$, and $t=t(u,v)$. We can represent this situation with the following tree diagram:

In the above tree diagram, 

we see that there are three different "levels" which represent the kinds of variables that we have. In particular, we have a dependent variable w, intermediate variables: x, y, z, and t, and independent variables u and v. A segment between two variables indicates that one of the variables is a function of the other. We can think of each segment as being labeled with a partial derivative. For example, the segment connecting w and x has the label $\frac{\partial w}{\partial x}$.

Now, in order to get a formula $\frac{\partial w}{\partial u}$ (or $\frac{\partial w}{\partial v}$) we consider every possible path from w to u (or from w to v). In doing this we obtain:

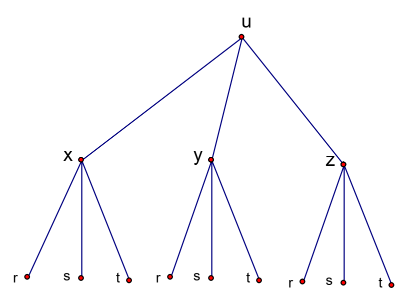
$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial u} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial u} + \frac{\partial w}{\partial t}\frac{\partial t}{\partial u}$$
$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial v} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial v} + \frac{\partial w}{\partial t}\frac{\partial t}{\partial v}$$

One should note that these are the formulas we would have obtained from the general chain rule.

Note: This idea of the tree diagram to represent the chain rule actually extends beyond the general chain rule discussed above. In particular, the tree diagram continues to work when we skip levels and/or have more than three levels!

<u>Example 4:</u> Suppose that $u = x^4 y + y^2 z^3$, $x = rse^t$, $y = rs^2 e^{-t}$, and $z = r^2 s\sin(t)$. Find the exact value of $\frac{\partial u}{\partial s}$ when $r = 2$, $s = 1$, and $t = 0$.

The tree diagram is:



So, by the chain rule we have that
$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial u}{\partial z}\frac{\partial z}{\partial s} = 4x^3 y\left(re^t\right) + \left(x^4 + 2yz^3\right)\left(2rse^{-t}\right) + 3y^2 z^2\left(r^2\sin(t)\right)$. We note that if $r = 2$, $s = 1$, and $t = 0$, then $x = 2$, $y = 2$, and $z = 0$. This means that

$$\left.\frac{\partial u}{\partial s}\right|_{r=2,\ s=1,\ t=0} = 4(2)^3(2)(2) + \left(2^4 + 0\right)\left(2(2)(1)e^{-0}\right) + 0 = 128 + 64 = 192 .$$